

Towards trustworthy machine fault diagnosis: A probabilistic Bayesian deep learning framework

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ABSTRACT

Fault diagnosis is efficient to improve the safety, reliability, and cost-effectiveness of industrial machinery. Deep learning has been extensively investigated in fault diagnosis, exhibiting state-of-the-art performance. However, since deep learning is inherently uninterpretable, the low trustworthiness of the diagnostic results given by these black-boxes has always been a limiting factor in industrial applications. Specially, the monitoring data under unforeseen domains will be easily misdiagnosed without any symptoms. To address this issue, this paper explores the fault diagnosis in a probabilistic Bayesian deep learning framework by exploiting an uncertainty-aware model to understand the unknown fault information and identify the inputs from unseen domains, ultimately achieving trustworthy diagnosis. Moreover, the diagnostic uncertainty is decomposed in two aspects: (1) epistemic uncertainty, reflecting the discrepancy of test input relative to the training data, and (2) aleatoric uncertainty, referring to the noise originating from the input, offering a deep understanding of the unknowns in the diagnostic model. The proposed framework not only can accurately identify the faults belonging to a known distribution, but also provides insights into uncertainty and avoid the erroneous decision-making. Last, but not least, comprehensive diagnostic experiments considering unseen scenarios are used to demonstrate the effectiveness of proposed framework, providing competitive results.

1. Introduction

Modern machinery equipment, such as gas turbines, aero-engines and wind turbines, are built to function in harsh and extreme conditions. Unexpected mechanical faults of such complex systems frequently occur during operation, posing a hazard to the safety of human life and property. Consequently, it is of significance to develop fault diagnosis and health management technologies for real-time surveillance of machine health condition [1–4]. This process is also known as intelligent fault diagnosis since the established model can automatically identify the machine faults and set an alarm for manual intervention. Intensive research on the advanced signal processing approaches, such as wavelet analysis, empirical mode decomposition (EMD) and sparse coding, has been conducted to extract the fault-sensitive features, such as the bearing fault feature frequency and gear mesh frequency [5–8]. These approaches can take advantage of the physical information and expert knowledge to make the accurate diagnosis of machine faults, whereas present a low automation degree. As the feature design and extraction need to be manually conducted, the diagnostic knowledge

hidden in the vast amounts of machine fault data cannot be automatically learned. Moreover, this feature extraction process requires the diagnostic experts to possess a wealth of prior knowledge and a deep understanding of the target machine. Sometimes, it is difficult to grasp the fault signatures and establish the general and efficient fault-sensitive features, due to the complex physical structure and fault mechanisms of the mechanical systems. All these disadvantages limit the flexible applications in different diagnostic tasks.

Among the numerous intelligent diagnosis methods developed over the past two decades, data-driven methods have significant advantages when lacking the physical information and internal fault mechanism of machine, and thus offer high usability and flexibility in a variety of industrial tasks. In recent years, there has been a surge in academic interest in developing deep learning-based methods in this sector [9–11]. Deep learning significantly reduces the reliance on hand-crafted feature engineering and expert knowledge, making it an efficient tool for fault diagnosis applications, especially with large-scale and multi-sensor machine data, thanks to an adaptive fault feature learning following

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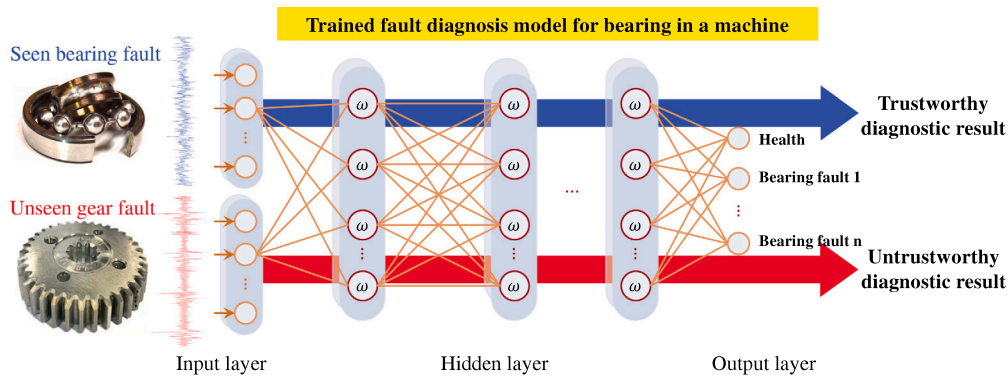


Fig. 1. The illustration of domain shift or OOD problem in deep learning-based machine fault diagnosis.

an end-to-end architecture. Stack auto-encoder (SAE), convolutional neural network (CNN), recurrent neural network (RNN), and other deep learning models, as well as their variation models, are researched for achieving state-of-the-art diagnostic performance [12–18]. Recent cutting-edge research focuses on the interpretability of deep neural networks, bridging the gap between physical signals and the diagnostic decision given by black-box models [19,20]. Wang et al. [19] presented a fully interpretable neural network to extract repetitive transients and monitor the machine health conditions, which significantly encourages the cooperation of signal processing-based physical features and neural networks. Li et al. [20] improved the traditional CNN with a continuous wavelet convolutional layer, and proposed a WaveletKernelNet to locate the fault-related impact components in the raw signal, which can clearly interpret the physical meaning of learned semantic information.

Despite a number of prominent successes, most of deep learning-based methods can only guarantee valid diagnostic results if the testing data has approximately the same distribution as the training data. Unfortunately, this requirement is difficult to satisfy in practical diagnosis applications. In the target domain where the trained deep model is deployed, the monitoring data are likely to differ from the source training data. The distribution discrepancy with respect to the known training data can be caused by unseen operating conditions such as varying loads, speeds, as well as unknown faults attributed to complex mechanical structures and failure mechanisms, and unseen data generation environments [21–23].

This challenging issue is often known as domain shift or out-of-distribution (OOD) problem [24–26]. When dealing with the unseen target data, deep models tend to make overconfident and untrustworthy diagnostic outputs without issuing any cautions. Fig. 1 illustrates the challenges of domain shift or OOD problem in deep learning. Given a well-trained deep model for diagnosing bearing health conditions, the unseen gear faults in target domain will be blindly diagnosed into any of the existing bearing faults. Such untrustworthy results fail to meet the reliability and safety requirements for machine fault diagnosis in safety-critical applications. As a result, there is an urgent need to address this issue. Minimizing the domain shift to make the trained model to generalize to target data has been a potential solution recently, and advances continue to be made.

To date, most fault diagnosis studies into domain shift minimization have focused on transfer learning [27–30]. Specially, as an important branch of transfer learning, domain adaptation can be useful in extracting domain-invariant features and bridging the gap of feature distribution across different domains, ensuring the generalization of the trained diagnostic model on the target data [31]. Thanks to its strong feature representation ability, the deep learning-assisted domain adaptation has led to a series of works dealing either with the fault diagnosis across different domains, such as different operating conditions and different machines [32–41], or with the open set diagnosis to isolate unseen target faults [42–45]. The first systematic study of transfer fault diagnosis between various operating conditions was

reported by Lu et al. [32]. The domain shift problem is considered, and the learned knowledge under source operating conditions are transferred and reused in other target operating conditions, including different loads and speeds for rotating machines. Michau et al. [33] utilized the adversarial training for domain adaptation among complementary operating conditions, and effectively detected anomaly in motor bearing data and turbofan engine simulation data. Most domain adaptation methods assume that machine health conditions are the same across domains, yet unseen faults are likely to occur, resulting in an open set scenario. In one of the earliest studies of open set diagnosis, Yang et al. [42] proposed a partial distribution adaptation sub-network to detect target samples with unknown faults during the transfer process, reducing the misdiagnosis rate. Zhang et al. [43] exploited the instance-level weighted domain adaptation to reject unseen target faults and learn domain-invariant features among the shared machine health conditions across domains.

The study on transfer learning is a positive step forward in addressing fault diagnosis in more realistic situations. However, there are two unresolved limitations on how to improve its application flexibility and ensure the trustworthiness of diagnostic results: (1) To perform domain adaptation, one must first gather the data distribution and properties in the target domain. In some cases, the desired target distribution may not be available due to data privacy requirements. Moreover, the specific target distribution can be uncertain at times. For instance, it is difficult to predict the target working load of wind turbines, since it varies with wind conditions. All these factors further hinder the application of knowledge transfer and domain adaptation in a flexible way. (2) The domain shift is minimized by comprehensively using both source and target data. The former source data are manually labeled, and thus are constrained to a known distribution. The latter target data are generally collected from a real-world application, and consist of a mass of healthy data and few fault data. In the complex application scenarios, the limited diagnostic knowledge gained from two domain data will be insufficient to cope with all possible operating conditions and machine health conditions. In other words, the transferred model is still subject to a certain distribution. The trustworthiness of the transfer diagnosis model's predictions remains unclear given any unseen inputs.

As a result, in addition to proactively reducing domain shift, it will be more rational and useful to develop a framework for effectively diagnosing machine faults and detecting unknowns for a trained model. That is, the fault diagnosis model should be concerned not only with accuracy but also with the trustworthiness of the diagnostic results. In doing so, the framework can alert human experts to probable false decisions during online surveillance, allowing for confident diagnostic results. It will be a valuable supplement and extension to the current framework and methodology, towards trustworthy machine fault diagnosis.

Naturally, uncertainty quantification is a promising method to evaluate what is unknown for a trained diagnosis model and when the diagnostic outputs can be trusted [46–48]. The higher uncertainty of

the outputs, the lower trustworthiness of the diagnostic model. Particularly, uncertainty in fault diagnosis can stem from various sources, each of which provides diagnostician with distinct information. The inherent randomness (or noise) in raw data that gives rise to the uncertainty is generally referred as aleatoric uncertainty, while the limited data and knowledge during training induce the inconsistency of outputs by varying model parameters, known as epistemic uncertainty [49–52]. Uncertainty is critical for building a trustworthy model and handling domain shift or OOD issues [53–56], yet it is often overlooked in machine fault diagnosis. Notably, most deep learning-based machine fault diagnosis only provides point-estimate predictions and cannot properly convey uncertainty in the diagnostic results. There are no systematic studies on how uncertainty quantification methods work on machine fault data, or on how to characterize internal changes of uncertainty with known or unknown fault data.

This paper proposes a novel uncertainty-aware machine fault diagnosis method in the probabilistic Bayesian deep learning framework. A comprehensive uncertainty treatment is achieved in twofold: (1) It follows Bayes' theorem to capture epistemic uncertainty by representing the network weights as a Gaussian distribution; and (2) It builds a probabilistic output layer that defines a multinomial distribution to represent the aleatoric uncertainty in the diagnosis of machine health condition. The proposed method effectively integrates the high-accuracy diagnosis and its uncertainty estimation, constituting a promising paradigm allowing the intelligent models to be used towards the trustworthy machine fault diagnosis. To the best of our knowledge, this is the first exploration to consider the trustworthiness of fault diagnostic results given by black-box models, which significantly enhances the reliability of current fault diagnosis methods in safety-critical applications. Specifically, this article makes four insights and contributions:

(1) The challenging untrustworthy fault diagnosis problem caused by domain shift or OOD is effectively addressed, providing a valuable supplement to the current fault diagnosis methodology.

(2) A novel uncertainty-aware machine fault diagnosis method is proposed in the probabilistic Bayesian deep learning framework. The uncertainty measures can be used to improve the model transparency, and inform us what is unknown for the fault diagnosis model and when the diagnostic results can be trusted.

(3) A uncertainty decomposition scheme is developed to analyze both aleatoric and epistemic uncertainty sources. The composition change in uncertainty is analyzed under three fault diagnosis scenarios, including unseen operating conditions, unseen faults and unseen noisy environments. Using the uncertainty decomposition helps us to understand and interpret the different facets of input fault data about whether fall outside the known distribution or are misdiagnosed due to the presence of inherent stochasticity and noise.

(4) Experiments on extensive machine fault data and diagnosis scenarios verify the effectiveness and superiority of the proposed trustworthy machine fault diagnosis method.

The reminder of this paper firstly introduces the preliminary in Section 2, then, describes the proposed method in Section 3, provides experiment analysis in Section 4, and finally concludes this article in Section 5.

2. Preliminary

2.1. Problem description

In this study, the uncertainty is communicated as a form of trustworthiness of intelligent fault diagnosis models. The motivation behind this idea is to establish a trustworthy machine fault diagnosis framework to deal with the challenging issues, i.e., domain shift or OOD problem. Let $\mathcal{D}$ denotes a domain consisting of the feature space $\mathcal{X}$ and the marginal distribution $P(X)$, where $X = \{x_i\}_{i=1}^N \in \mathcal{X}$ is the collection of samples. Before training a fault diagnosis model, a source domain

$\mathcal{D}_s = \{\mathcal{X}_s, P_s(X)\}$, also referred as seen domain, is assumed to be available. $\mathcal{D} = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ is a collection of labeled samples in $\mathcal{D}_s$. The predictive function $f_s(\cdot)$ of fault diagnosis model is trained in $\mathcal{D}$. The $f_s(\cdot)$ can automatically identify the machine health condition by mapping the samples in $\mathcal{X}_s$ to the label space $\mathcal{Y}_s$.

After training, the fault diagnosis model is deployed in practical industrial scenarios, denoted as target domain $\mathcal{D}_t$, also referred as unseen domain. The domain shift or OOD problem implies $\mathcal{X}_s \neq \mathcal{X}_t$, $P_s(X) \neq P_t(X)$, $\mathcal{Y}_s \neq \mathcal{Y}_t$ or $f_s(\cdot) \neq f_t(\cdot)$. These discrepancies between the $\mathcal{D}_s$ and $\mathcal{D}_t$ are ubiquitous but difficult to solve in practice. In this case, the predictive function $f_s(\cdot)$ of trained diagnosis model is likely to misdiagnose the health condition of samples $\mathcal{D} = \{(x_i^t, y_i^t)\}_{i=1}^{N_t}$ in $\mathcal{D}_t$, and gives the untrustworthy decisions [57]. To form a trustworthy diagnosis framework in safety-critical applications, this study focuses on not only the diagnostic decision of the models, but also how confident the decision is. The predictive uncertainty $\mathbb{H}[\hat{y}_i^t | x_i^t, \mathcal{D}]$ of the $f_s(\cdot)$ will be investigated to emit warnings for the potential false diagnosis. The $\hat{y}_i^t$ is the predicted machine health condition by the trained model.

2.2. Probabilistic Bayesian deep learning

In conventional deep neural networks (DNN), the learning process involves optimizing a loss function to find the optimal model parameters with respect to the known training data. In classification tasks, the cross-entropy loss function is most commonly used in conjunction with the well-known optimization techniques (e.g. gradient descent). From a probabilistic perspective, the model parameters w are obtained or trained in the manner of maximum likelihood estimation (MLE), that is,

$$\begin{aligned} w^* &= \arg \max_w p(\mathcal{D} | w) \\ &= \arg \max_w \sum_i \log p(y_i | x_i, w). \end{aligned} \quad (1)$$

Actually, the trained model parameters, i.e., the weights and biases, are restricted to a point estimate in conventional DNN, resulting in deterministic outputs for any inputs. Now in Bayesian context, instead of giving a point estimate for each parameter, the probability distribution is provided across all parameters. This is referred as the posterior distribution $p(w | \mathcal{D})$. The posterior distribution helps Bayesian neural network to capture the uncertainty of network weights during the classification or regression [58]. Fig. 2 illustrates the details of a Bayesian CNN, which utilizes a weight distribution rather than the point estimate in conventional CNN.

To find the posterior distribution over parameters, the simple rules of probability, i.e., Bayes' rule need to be used here:

$$p(w | \mathcal{D}) = \frac{p(\mathcal{D} | w)p(w)}{p(\mathcal{D})} = \frac{p(\mathcal{D} | w)p(w)}{\int_w p(\mathcal{D} | w)p(w)dw}, \quad (2)$$

where $p(\mathcal{D} | w)$ represents the likelihood of data for a specific weight collection w , $p(w)$ is the prior distribution, and $p(\mathcal{D})$ is the marginal likelihood. The Gaussian distribution is used as the prior in this study. In the Bayesian framework, the learning process entails shifting the belief about the weights from the prior to the posterior as a result of observing the data.

Given a specific setting of w , both of the $p(\mathcal{D} | w)$ and $p(w)$ are tractable. However, the calculation of the marginal likelihood $p(\mathcal{D})$ is difficult, since it requires the marginalization over all parameter settings: $\int_w p(\mathcal{D} | w)p(w)dw$. This becomes even more challenging in DNNs due to the large number of parameters. To address this, an approximation is required to evaluate the posterior, i.e., variational inference.

2.3. Variational inference

The goal of variational inference is to create a variational distribution $q_\theta(w)$, also referred as approximate distribution, that is as close to

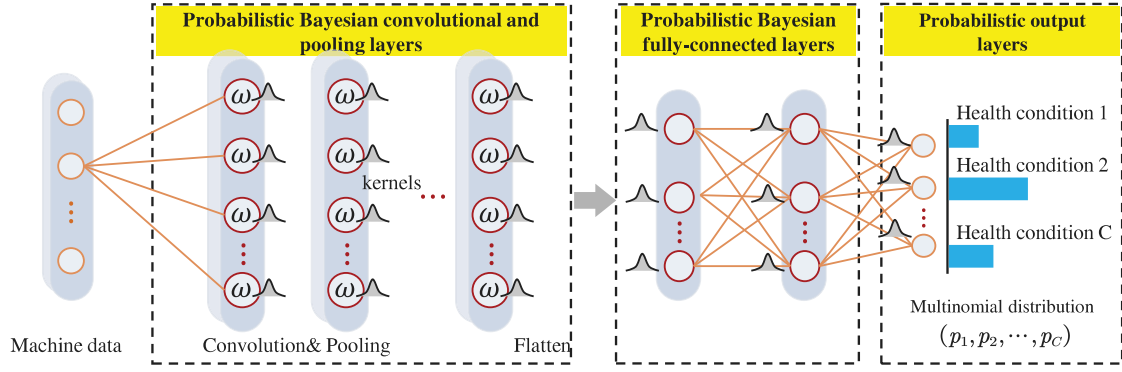


Fig. 2. Probabilistic Bayesian convolutional neural network.

the genuine posterior $p(w|D)$ as possible. Generally, the $q_\theta(w)$ is defined by a simpler and tractable family of distributions. To achieve this approximation, the difference between two probability distributions $q(x)$ and $p(x)$ should be estimated by Kullback–Leibler (KL) divergence, defined as:

$$\begin{aligned} \text{KL}[q(x) \parallel p(x)] &= \mathbb{E}_{q(x)} \left[\log \frac{q(x)}{p(x)} \right] \\ &= \int q(x) \log \frac{q(x)}{p(x)} dx. \end{aligned} \quad (3)$$

Then, the minimization of KL divergence between variational distribution $q_\theta(w)$ and the true posterior distribution $p(w|D)$ yields:

$$\begin{aligned} \text{KL}[q_\theta(w) \parallel p(w|D)] &= \mathbb{E}_{q_\theta(w)} \left[\log \frac{q_\theta(w)}{p(w|D)} \right] \\ &= \int q_\theta(w) \log \frac{q_\theta(w)}{p(w|D)} dw \\ &= \int q_\theta(w) \log \frac{q_\theta(w)p(D)}{p(D)p(w)} dw \\ &= \int q_\theta(w) \log \frac{q_\theta(w)}{p(w)} dw - \int q_\theta(w) \log p(D) dw \\ &\quad + \int q_\theta(w) \log p(D) dw \\ &= \text{KL}[q_\theta(w) \parallel p(w)] \\ &\quad - \mathbb{E}_{q_\theta(w)} [\log p(D|w)] + \log p(D). \end{aligned} \quad (4)$$

In Eq. (4), the $\text{KL}[q_\theta(w) \parallel p(w)]$ is referred as complexity cost, which denotes the similarity between the variational distribution $q_\theta(w)$ and the prior $p(w)$. The $\mathbb{E}_{q_\theta(w)} [\log p(D|w)]$ is called likelihood cost, which describes how well the model is fit to the data. The $\log p(D)$ is a constant. As a result, the minimization of $\text{KL}[q_\theta(w) \parallel p(w|D)]$ can be translated into the optimization of maximizing the evidence lower bound (ELBO):

$$\begin{aligned} \mathcal{L}_{ELBO}(D, \theta) &= \mathbb{E}_{q_\theta(w)} [\log p(D|w)] - \text{KL}[q_\theta(w) \parallel p(w)] \\ &= \mathbb{E}_{q_\theta(w)} [\log p(D|w)] - \mathbb{E}_{q_\theta(w)} [\log q_\theta(w)] + \mathbb{E}_{q_\theta(w)} [\log p(w)]. \end{aligned} \quad (5)$$

For the sake of computational complexity, the ELBO is approximated by an unbiased Monte-Carlo estimate by drawing samples from the variational posterior distribution $q_\theta(w)$. Notably, Flipout method is a well-established implementation that is scalable and efficient to maintain a low variance of estimates [59]. Here, the tractable optimization object, i.e., maximization w.r.t. θ , becomes:

$$\mathcal{L}_{ELBO}(D, \theta) \approx \sum_{k=1}^K \log p(D|w^{(k)}) - \log q_\theta(w^{(k)}) + \log p(w^{(k)}). \quad (6)$$

where K denotes the number of samples from $q_\theta(w)$. The backpropagation algorithm in conventional DNN can be utilized to optimize the variational parameter θ and help the learning in Bayesian deep learning.

3. Proposed method

3.1. Trustworthy machine fault diagnosis in probabilistic Bayesian framework

In this study, a uncertainty-aware method is explored in the probabilistic Bayesian deep learning framework towards the trustworthy machine fault diagnosis. Specifically, the probabilistic Bayesian CNN is used as the backbone model. Note that the proposed method is not dependent upon a particular deep learning model. It is flexible to transfer the diagnostic procedure to other types of models, such as feedforward DNN and RNN, in Bayesian framework. The overall procedure of the proposed method is illustrated in Fig. 3. The general steps are summarized as follows:

(1) The raw data of monitored machine are collected in a source domain D_s (seen domain), where the specific machine health conditions and operating conditions are known and pre-defined. By partitioning and labeling the raw data, the samples collection $D = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ is prepared for training the diagnostic model.

(2) The probabilistic Bayesian CNN is learned with variational inference. The posterior parameter distribution is approximated to the variational distribution with the observation of $D = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$.

(3) After training, the learned probabilistic Bayesian CNN is deployed in real applications, i.e., target domain D_t (unseen domain). Sometimes, due to the domain shift and OOD problem, discrepancies exist between D_s and D_t . Both of the data $D = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ in D_s and $D = \{(x_i^t, y_i^t)\}_{i=1}^{N_t}$ in D_t are likely to be monitored.

(4) For each monitoring sample, the machine health condition can be automatically diagnosed by the learned probabilistic Bayesian CNN, whose details will be described in Section 3.2.

(5) To evaluate the trustworthiness of the decisions given by the diagnostic model, the uncertainty estimation is conducted. Uncertainty decomposition with respect to aleatoric and epistemic uncertainty is further analyzed, so as to explain the different facets of unknowns in the diagnostic model. The details will be described in Section 3.2. and Section 3.3.

(6) Additional human intervention is necessary to the samples with abnormal signs on uncertainty estimation in safety-critical applications. The uncertainty estimation and decomposition results will be combined with human experts' domain knowledge to uncover the reason of potential misdiagnosis, such as unseen faults, unseen operating conditions, and unseen noisy environments.

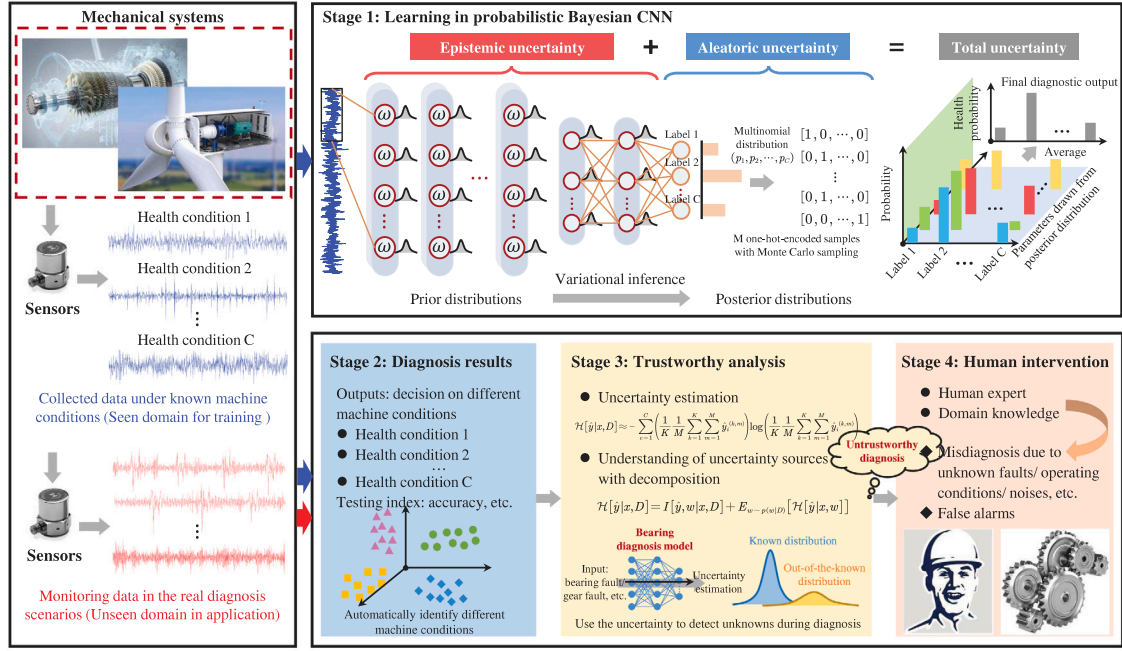


Fig. 3. Illustration of the trustworthy machine fault diagnosis in probabilistic Bayesian framework.

3.2. Uncertainty-aware trustworthy analysis

Given a testing sample (x_i, y_i) , the learned variational distribution $q_\theta(w)$ is used to approximate the actual posterior distribution of the parameters for multinomial distribution.

$$p(\hat{y}_i | x_i, D) \approx \int p(\hat{y}_i | x_i, w) q_\theta(w) dw. \quad (7)$$

Suppose the number of machine health conditions that need to be diagnosed in seen domain is C , the multinomial distribution parameters with respect to weight sample $w^{(k)}$ are denoted as:

$$O = \begin{bmatrix} p(\hat{y}_i = 1 | x_i, w^{(k)}) \\ p(\hat{y}_i = 2 | x_i, w^{(k)}) \\ \vdots \\ p(\hat{y}_i = C | x_i, w^{(k)}) \end{bmatrix} = \frac{1}{\sum_{j=1}^C \exp(o_j)} \begin{bmatrix} \exp(o_1) \\ \exp(o_2) \\ \vdots \\ \exp(o_C) \end{bmatrix}, \quad (8)$$

where o_j is the output of the j th output neuron.

To avoid the integration over the weights, we resort to approximations by two-fold Monte Carlo sampling. First, MC sampling is utilized to take K samples from the weight distribution $w^{(k)} \sim q_\theta(w)$. One can obtain the multinomial distribution $p(\hat{y}_i | x_i, w^{(k)})$ for each weight set $w^{(k)}$. Second, use the multinomial distribution to generate M one-hot-encoded samples denoted by $\hat{y}_i^{(k,m)}$. All those samples are aggregated to compute the final mean prediction. One could assign the machine to a health condition which has the highest mean of the predicted probabilities.

$$\begin{aligned} p(\hat{y}_i | x_i, D) &\approx \frac{1}{K} \sum_{k=1}^K p(\hat{y}_i | x_i, w^{(k)}) \\ &\approx \frac{1}{K} \frac{1}{M} \sum_{k=1}^K \sum_{m=1}^M \hat{y}_i^{(k,m)}. \end{aligned} \quad (9)$$

For the specific weight sample $w^{(k)} \sim q_\theta(w)$, a confident output gives the majority of probability mass on a single machine health condition, whereas the unconfident one tends to place equal probability mass on different machine health conditions. The former will induce the low entropy, while the latter will produce the high entropy. Therefore, con-

sidering the weight distribution, the uncertainty of diagnostic result can be effectively characterized by the entropy in the output distribution:

$$\begin{aligned} \mathbb{H}[\hat{y}_i | x_i, D] &= - \sum_{c=1}^C p(\hat{y}_i = c | x_i, D) \times \log p(\hat{y}_i = c | x_i, D) \\ &= - \sum_{c=1}^C \left(\int p(\hat{y}_i = c | x_i, w) p(w | D) dw \right) \\ &\quad \times \log \left(\int p(\hat{y}_i = c | x_i, w) p(w | D) dw \right) \\ &\approx - \sum_{c=1}^C \left(\frac{1}{K} \sum_{k=1}^K p(\hat{y}_i = c | x_i, w^{(k)}) \right) \\ &\quad \times \log \left(\frac{1}{K} \sum_{k=1}^K p(\hat{y}_i = c | x_i, w^{(k)}) \right) \\ &\approx - \sum_{c=1}^C \left(\frac{1}{K} \frac{1}{M} \sum_{k=1}^K \sum_{m=1}^M \hat{y}_i^{(k,m)} \right) \log \left(\frac{1}{K} \frac{1}{M} \sum_{k=1}^K \sum_{m=1}^M \hat{y}_i^{(k,m)} \right). \end{aligned} \quad (10)$$

Note that in Eq. (10), when the posterior weight distribution $q_\theta(w)$ becomes a point, the Bayesian deep learning model becomes the conventional DNNs. The point estimate in DNNs could yield very confident, yet overconfident decision, suggesting a lack of the ability to describe the inherent uncertainty. Using different weight samples from the posterior weight distribution $w^{(k)} \sim q_\theta(w)$ could lead to the disagreement in their decisions, which can be used to characterize the certainty of unknowns in the diagnostic result [60].

Uncertainty enables to reflect how confident the model is when diagnosing the specific machine data. High uncertainty corresponds to unknowns in the input data fed to the trained model, and the diagnostic decisions given by the model is untrustworthy. Conversely, low uncertainty indicates the model has sufficient knowledge to deal with the machine data. The uncertainty estimation allows the human experts to have a better understanding of the automatic model outputs, and further integrate the model's diagnostic decisions with the experts' judgement.

3.3. Understanding of uncertainty sources

Further decomposing the total uncertainty in the prediction can provide additional insights into the machine fault diagnosis. Particularly, the total uncertainty is the sum of aleatoric and epistemic uncertainty as shown in Eq. (11) [49,50]. The first term represents the epistemic uncertainty and is defined as the amount of information we would gain about the model parameters if we were received a new labeled samples given the dataset [61]. The second term denotes the aleatoric uncertainty and is measured as the average entropy of the prediction given data input x_i and fixed weights drawn from the posterior distribution $w^{(k)} \sim p(w|D)$.

$$\mathbb{H}[\hat{y}_i|x_i, D] = \mathbb{I}[\hat{y}_i, w|x_i, D] + \mathbb{E}_{w \sim p(w|D)} [\mathbb{H}[\hat{y}_i|x_i, w]]. \quad (11)$$

Aleatoric uncertainty, refers to uncertainty inherent in the input samples, which can be generated by a variety of unobserved factors such as sensor faults and changing environmental conditions. In the context of fault diagnosis, aleatoric uncertainty is mathematically represented by the multinomial distribution $p(\hat{y}_i|x_i, w^{(k)})$ given the input instance x_i and fixed network weight parameters $w^{(k)}$:

$$\begin{aligned} \mathbb{E}_{w \sim p(w|D)} [\mathbb{H}[\hat{y}_i|x_i, w]] \\ &= - \int p(w|D) \left[\sum_{c=1}^C p(y_i = c|x_i, w) \log p(\hat{y}_i = c|x_i, w) \right] dw \\ &\approx - \int q_\theta(w) \left[\sum_{c=1}^C p(\hat{y}_i = c|x_i, w) \log p(\hat{y}_i = c|x_i, w) \right] dw \\ &\approx - \frac{1}{K} \sum_{k=1}^K \sum_{c=1}^C p(\hat{y}_i = c|x_i, w^{(k)}) \log p(\hat{y}_i = c|x_i, w^{(k)}) \\ &\approx - \frac{1}{K} \sum_{k=1}^K \sum_{c=1}^C \left(\frac{1}{M} \sum_{m=1}^M \hat{y}_i^{(k,m)} \right) \log \left(\frac{1}{M} \sum_{m=1}^M \hat{y}_i^{(k,m)} \right). \end{aligned} \quad (12)$$

Epistemic uncertainty, refers to the uncertainty in the model parameters, caused by a lack of knowledge about machine, such as operating conditions and fault phenomena. The posterior weight distribution $p(w|D)$ mathematically represents epistemic uncertainty. It can be calculated as the difference between Eqs. (10) and (12):

$$\begin{aligned} \mathbb{I}[\hat{y}_i, w|x_i, D] &= \mathbb{H}[\hat{y}_i|x_i, D] - \mathbb{E}_{w \sim p(w|D)} [\mathbb{H}[\hat{y}_i|x_i, w]] \\ &\approx - \sum_{c=1}^C \left(\frac{1}{K} \frac{1}{M} \sum_{k=1}^K \sum_{m=1}^M \hat{y}_i^{(k,m)} \right) \log \left(\frac{1}{K} \frac{1}{M} \sum_{k=1}^K \sum_{m=1}^M \hat{y}_i^{(k,m)} \right) \\ &\quad + \frac{1}{K} \sum_{k=1}^K \sum_{c=1}^C \left(\frac{1}{M} \sum_{m=1}^M \hat{y}_i^{(k,m)} \right) \log \left(\frac{1}{M} \sum_{m=1}^M \hat{y}_i^{(k,m)} \right). \end{aligned} \quad (13)$$

High aleatoric uncertainty means that the available data is noisy and is insufficient to identify the machine's health condition. Hence, more information should be gathered to refine the current data, such as installing better sensors or improving calibration procedures. High epistemic uncertainty means we only have an incomplete knowledge of the machine. Therefore, more effort should be made to improve our understanding of the machine, such as planning new experiments and/or simulations, as well as gathering more helpful data on other conceivable operating conditions or fault phenomena.

4. Experiment analysis

In this section, a rotor bearing fault diagnosis case is used to validate the effectiveness of the proposed method. We demonstrate the robust performance of our proposed method in dealing with seen domain and three possible scenarios of unseen domain in practical fault diagnosis, including unseen fault, varying operating condition and noisy environment. In each of the scenario, we aim to answer three questions: (1) Can the proposed method diagnose seen domain as comparable as

the benchmarking deep learning models? (2) How does the proposed method work to identify the unseen domain? How effective is the proposed method to recognize the unseen domain and make trustworthy diagnosis? (3) Can we use the uncertainty decomposition (i.e., aleatoric and epistemic uncertainties) to help understand the possible sources of unseen domain in machine fault diagnosis?

4.1. Bearing fault data descriptions

The bearing fault data is collected from a HD-FD-03X rotor fault test rig. The experimental setup is shown in Fig. 4. The rotor is driven by a motor, and the load at the output end is provided by a brake loader. The acceleration signals on the bearing pedestal of vertical direction are to diagnose bearing health conditions. The sampling frequency is 20 kHz. The signal is divided into segments (each with 2048 data points) that do not overlap.

In the seen domain (where the diagnostic model is trained), the motor speeds are 2000, 2250, 2500, 2750 and 3000 r/min, with a load of 0 N. The experiments were conducted to identify four different bearing health states, namely, healthy (H), out race fault (OF), inner race fault (IF) and rolling element fault (RF). For each health condition, a total of 500 samples are used for training, and another 100 samples are used for testing. The selected samples are evenly distributed in each of the five different rotating speeds.

In the unseen domain, three scenarios are considered, including unseen faults, unseen operating conditions and unseen noisy environments. (1) The gearbox fault (gear tooth broken) is considered as an unseen fault, since the model is only trained on bearing fault data. There are 400 gearbox fault samples for testing. The motor speeds and data collecting settings are identical to those in the seen domain. (2) The motor speeds are adjusted to 1750, 1500, 1250 and 1000 r/min, which mimics the possible unseen operating conditions. (3) Gaussian noise is added to the testing samples of the seen domain. This generates 400 noisy samples to mimic unseen noisy environments. With a 15 dB step interval, the signal-to-noise ratio (SNR) varies from 40 dB to -50 dB. This settings aim to simulate the environmental noise and measurement error, which may compromise the reliability of fault diagnosis. The details of experiment settings are listed in Table 1.

4.2. Implementation details

The probabilistic Bayesian CNN is used as the backbone model in this study. The network architecture is similar to the commonly used one-dimensional CNN, while each parameter is represented as a Gaussian distribution. Five Bayesian convolutional layers are considered in the probabilistic Bayesian CNN, whose kernel sizes are $\{32 \times 1, 16 \times 1, 8 \times 1, 3 \times 1, 2 \times 1\}$, respectively, and the kernel numbers are $\{16, 16, 32, 32, 64\}$, respectively. A max-pooling layer with the pooling size of 2×1 is appended to each Bayesian convolutional layer. After the convolution and pooling operations, the extracted features are flattened as the input to Bayesian fully-connected layers. Three Bayesian fully-connected layers are designed, including two hidden layers and an output probabilistic layer. The numbers of neurons in the two hidden layers are 128 and 64, respectively. The number of neurons in output probabilistic layer is 4, representing the four different bearing conditions.

The probabilistic Bayesian CNN is learned in Tensorflow 2.2.0 environment. Flipout method is executed to train the Bayesian layers and help the approximation of posterior distribution during variational inference. The optimizer is SGD with the learning rate of 0.001. The batch size is 32. The early stopping with patience 10 is used as the end criterion during training.

Specifically, we benchmark our proposed method with two popular deep learning models including: one-dimensional CNN [23] and deep residual network (ResNet) [62]. The network parameters of CNN are the same as the ones of probabilistic Bayesian CNN. The only difference between them is that each parameter is a deterministic value in CNN, while a distribution in probabilistic Bayesian CNN. For the ResNet, the designed ResNet18 structure by He et al. [62] are used for comparison, whose convolutional kernels are also set to one-dimension.

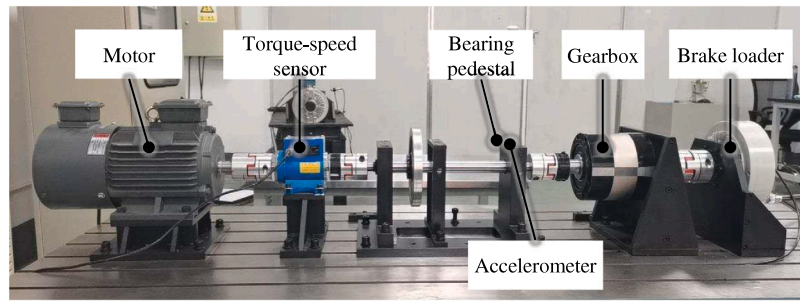


Fig. 4. The HD-FD-03X rotor fault test rig.

Table 1
Descriptions of experiment settings in both seen and unseen domains.

Domain	Health condition	Labels	Operating conditions	Number of training samples	Number of testing samples
Seen domain	H	0	2000, 2250, 2500,	4 × 500	4 × 100
	OF	1	2750, 3000 r/min		
	IF	2			
	RF	3			
Unseen fault	Gear fault	\	2000, 2250, 2500, 2750, 3000 r/min	\	1 × 400
Unseen operating conditions	H	0	1000, 1250, 1500,	\	4 × 100
	OF	1	1750 r/min		
	IF	2			
	RF	3			
Unseen noise environments	H	0	2000, 2250, 2500,	\	4 × 100
	OF	1	2750, 3000 r/min		
	IF	2			
	RF	3			

4.3. Results analysis for unseen faults

This section examines the performance of our proposed method from two aspects: a comparable performance with the traditional diagnosis in handling seen domain; and a unique capability in recognizing unseen domains. It is observed that all three models achieve high-accuracy results in diagnosing seen domain. The diagnostic accuracies of CNN, ResNet and probabilistic Bayesian CNN are 98.8%, 98% and 98.5%, respectively. This implies that our proposed method also performs satisfactorily in diagnosing faults within the seen domain. On the other hand, when dealing with unseen fault, CNN and ResNet undoubtedly give incorrect results, since the unseen gear fault is not used to develop the trained models. This highlights the necessity and advantages of our proposed probabilistic Bayesian CNN. In this context, we discuss the our proposed method by an illustrative example, total uncertainty analysis and uncertainty decomposition.

Fig. 5 presents an example for each health condition within the seen domain and unseen fault, and then shows the predictions based on CNN, ResNet and probabilistic Bayesian CNN. As expected, CNN and ResNet are deterministic and can only provide a point estimate of the probability for each health condition, while our proposed probabilistic Bayesian CNN can provide the predictive uncertainty for each health condition. As discussed in Section 3.2, the two-fold Monte Carlo sampling is followed to characterize both epistemic and aleatoric uncertainties. Then obtain a distribution over each machine health condition, as illustrated by the violin plot. When handling the samples of the seen domain in Fig. 5(a)–(d), all three models can offer correct diagnostic results. Moreover, probabilistic Bayesian CNN not only makes the correct predictions, but also shows its high confidence as reflected by the small uncertainties. Fig. 5(e) presents the results of diagnosing unseen gear fault. CNN and ResNet confidently misdiagnose it as predefined RF and IF respectively, without emitting any warnings. Interestingly, in the results of probabilistic Bayesian CNN, there are large uncertainties for the probability regarding each health condition.

This implies that the diagnostic results are untrustworthy and should require further investigations.

To quantify the total predictive uncertainty, the entropy of the probability distribution is calculated in probabilistic Bayesian CNN. The predictive uncertainty of testing samples in both seen domain and unseen fault is exhibited in Fig. 6(a). Fig. 6(b) shows a comparison of the uncertainty distribution in the seen domain and unseen fault. Clearly, the uncertainty of unseen fault is much higher than that of seen domain. In other words, probabilistic Bayesian CNN is confident in making correct diagnosis with a small uncertainty, and recognizing the unseen faults by showing a substantially large uncertainty. This implies that the uncertainty conveyed by probabilistic Bayesian CNN is sensitive to the unseen fault and can be an effective measure to improve trustworthy of fault diagnosis.

We further decompose the predictive uncertainty into aleatoric and epistemic uncertainties as shown in Fig. 6(c)–(d). In the seen domain, the aleatoric and epistemic uncertainties are similar, indicating that the two sources of uncertainty have approximately equal contributions to the total uncertainty. When diagnosing unseen fault, both sources of uncertainty increase as the total uncertainty increases. More importantly, compared with aleatoric uncertainty, epistemic uncertainty makes a more significant contribution to the total uncertainty. This intriguing result can be explained by the fact that the unseen gear fault is out of the known faults, and such insufficient diagnostic knowledge will give rise to epistemic uncertainty. The decomposed results provide more insights that is obscured by sole checking the total uncertainty. More analysis and discussions about this point will be given in Section 4.6.

4.4. Results analysis for unseen operating conditions

In real-world applications, machines often function under a variety of operating conditions, including unseen or unpredictable speeds and loads. A set of diagnosis experiments for unseen operating conditions are listed in Table 1. Fig. 7 displays the time waveforms of testing

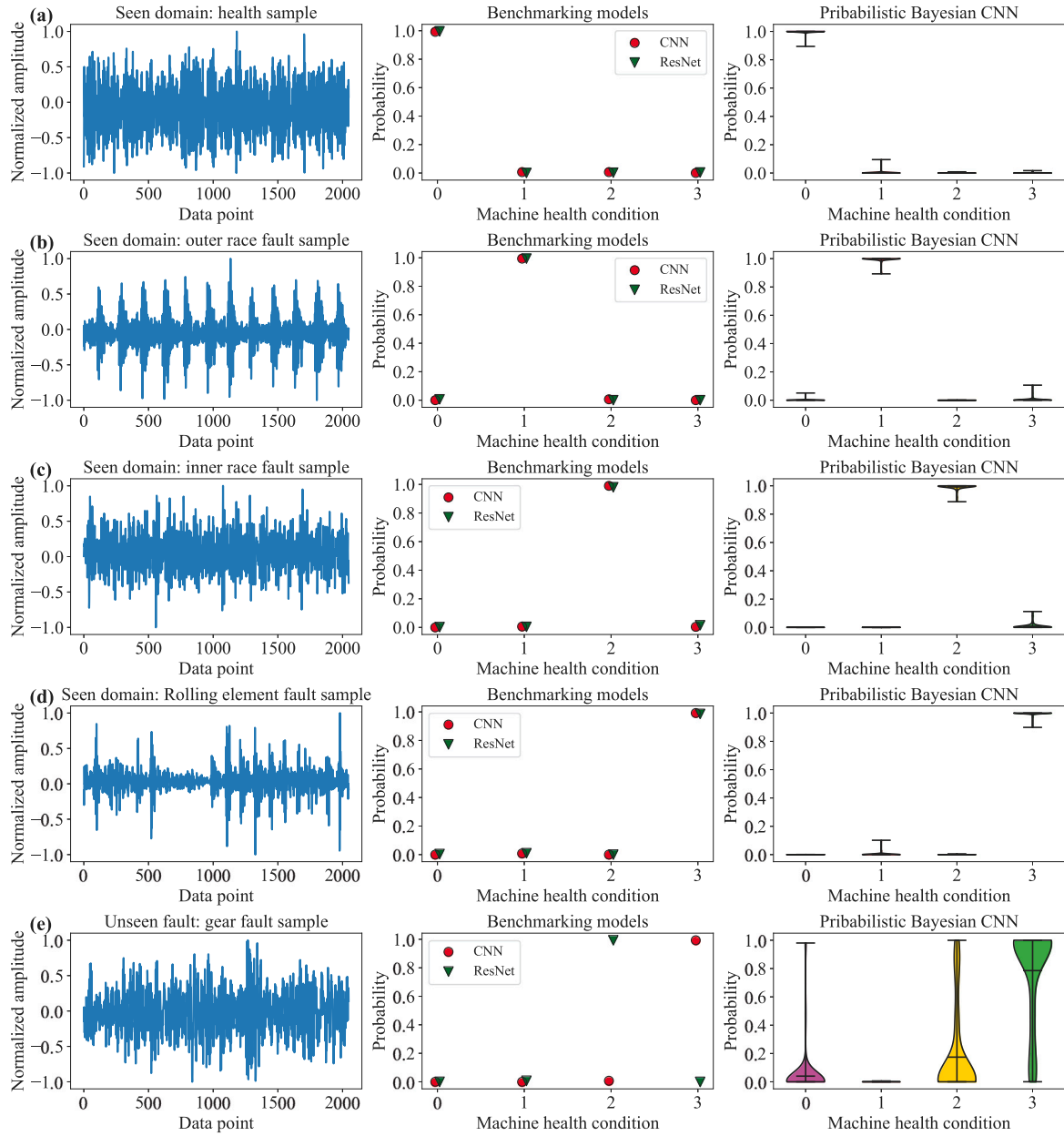


Fig. 5. The diagnosis of the testing samples of seen domain and unseen fault with benchmarking models and probabilistic Bayesian CNN. (a) The original time waveform of each sample. (b) Diagnostic results of benchmarking models trained in the seen domain. (c) A distribution over diagnostic results of probabilistic Bayesian CNN (also trained in the seen domain).

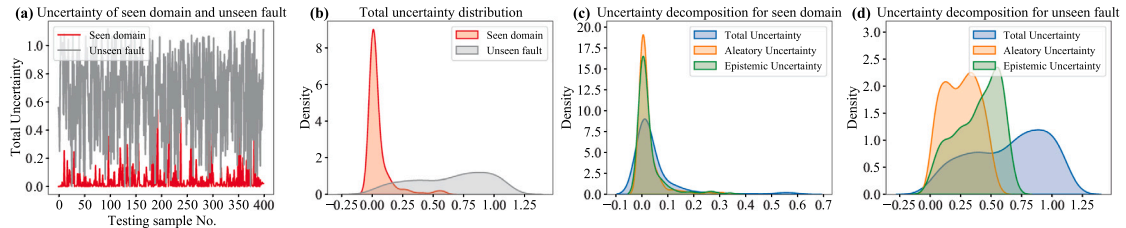


Fig. 6. Uncertainty estimation for the testing samples of seen domain and unseen fault. (a) The total uncertainty of testing samples. (b) The distribution of total uncertainty. (c)–(d) The distribution of decomposed uncertainty in seen domain and unseen fault, respectively.

samples in seen domain and unseen operating conditions, respectively. It can be seen that the fault signatures change with rotating speeds. For example, in the outer race fault, the number of periodicity impacts within a fixed time window is closely related to the rotating speed. With

the decrease of rotating speed, the number of impacts also decreases, leading to a distinct domain shift between seen domain and unseen operating conditions. Note that the greater the discrepancy between the two speeds, the larger the distribution shift between the two domains.

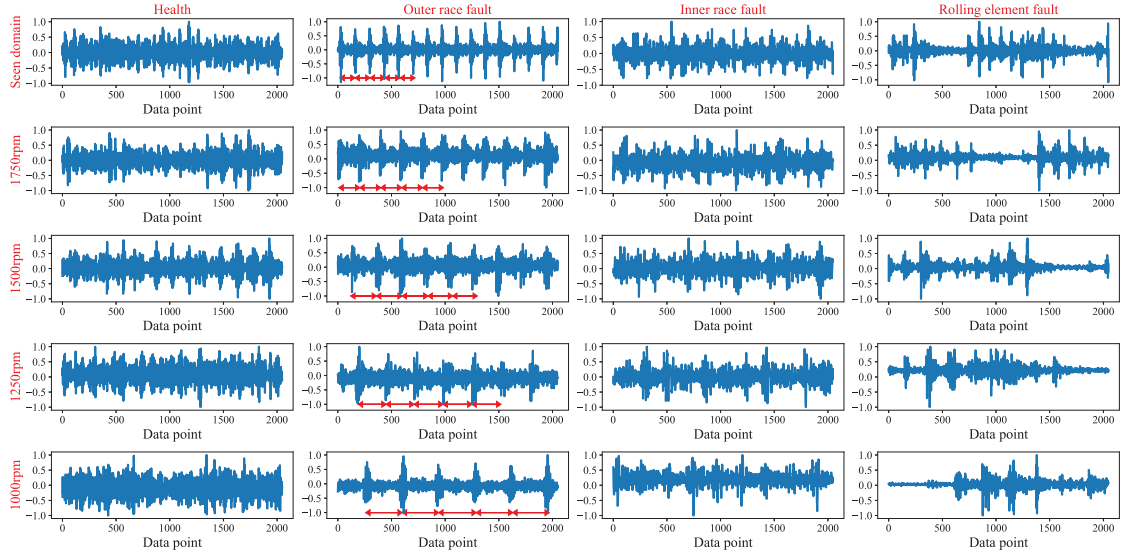


Fig. 7. The time waveforms of the testing samples in seen domain and unseen operating conditions.

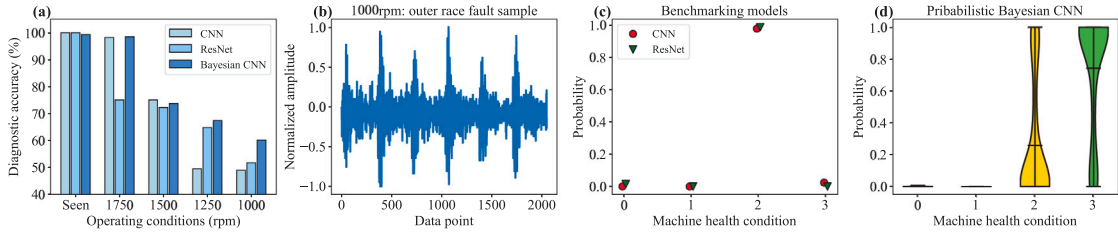


Fig. 8. The diagnosis of the testing samples of unseen operating conditions with benchmarking models and probabilistic Bayesian CNN. (a) Overall diagnostic accuracies under different operating conditions. (b) The original time waveform of a outer race fault sample under 1000 r/min. (c)–(d) Diagnostic results of benchmarking models and probabilistic Bayesian CNN for the outer race fault sample.

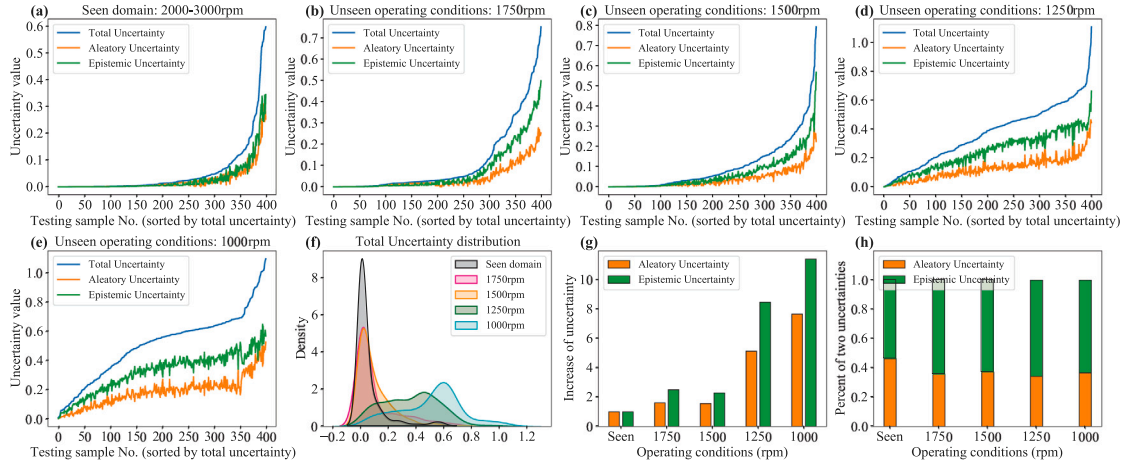


Fig. 9. Uncertainty estimation and decomposition for the testing samples of seen domain and different operating conditions. (a)–(e) The total uncertainty and decomposed uncertainty of testing samples in different domains. (f) The distribution of total uncertainty in different domains. (g)–(h) Analysis of uncertainty composition changes with respect to different domains.

The overall diagnostic accuracies of 400 testing samples are shown in Fig. 8(a). All models achieve high-accuracy diagnostic results in the seen domain, whereas have poor performance in unseen operating conditions due to the domain shift, especially in the operating conditions of 1000 r/min and 1250 r/min. Moreover, Fig. 8(c)–(d) gives the detailed diagnostic results of a outer race fault sample. Similar to Section 4.3, the benchmarking models overconfidently make false

diagnostic decision; the large uncertainty conveyed by probabilistic Bayesian CNN helps recognize the unseen operating conditions.

Fig. 9 summarizes the uncertainty estimation and decomposition results for both seen domain and unseen operating conditions. This allows one to examine the trustworthiness of the diagnostic results using probabilistic Bayesian CNN from two aspects: (1) can we use total uncertainty to distinguish the unseen scenarios with rotating speeds

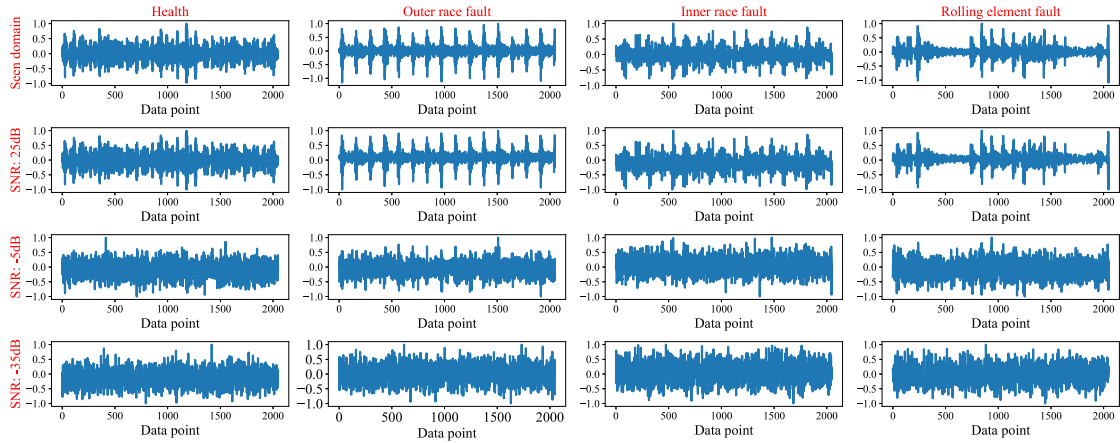


Fig. 10. The time waveforms of the testing samples in seen domain and unseen noisy environments.

different from the seen domain? and (2) How are the various rotating speeds reflected by the aleatoric and epistemic uncertainty sources?

In Fig. 9(a), the total uncertainty is close to 0 for most of testing samples in seen domain, indicating the trustworthy diagnostic results. From Fig. 9(b)–(e), the total uncertainty continues to grow as the rotation speed increases further away from the seen domain (2000–3000 r/min). This can also be supported by the discrepancy between uncertainty distributions as shown in Fig. 9(f). This implies that the total uncertainty follows an increasing trend as a function of the difference between rotating speed in the seen domain and unseen domain. Despite the fact that the first two unseen operating conditions (i.e., 1750 and 1500 r/min) and the seen domain have a lot of overlap in their uncertainty distributions, it is still clear to spot the differences, notably between the seen domain and the unseen operating condition of 1500 r/min. Overall, the uncertainty estimation can help human experts better understand the model's trustworthiness when dealing with unseen operating conditions, especially for those that are much different from the seen domain (such as 1000 and 1250 r/min). Note the diagnostic accuracies of the unseen operating conditions (1500, 1250 and 1000 r/min) are only 73.7%, 67.3% and 60.2% respectively. This is a significant evidence of the untrustworthy diagnosis of a number of testing samples under these operating conditions.

According to Fig. 9(b)–(e), both aleatoric and epistemic uncertainties also follow an increasing trend with a larger difference between rotating speed in seen domain and unseen domain. Another important finding is that epistemic uncertainty increases more dramatically than aleatoric uncertainty, based on examining the degree of changes in their average value in different operating condition. Specifically, we calculate the average aleatoric and epistemic uncertainties in the seen domain and the other three unseen operating conditions. Let us suppose that we use the average value in the seen domain as a reference value (represented by 1). Then we can examine the changes of uncertainty in the varying operating condition, by calculating the ratio of average value of the unseen operation condition to the reference value as presented in Fig. 9(f). Furthermore, we show the proportions of the two sources of uncertainty in the total uncertainty in Fig. 9(h). All these results suggest that epistemic uncertainty accounts for a larger proportion of total uncertainty. This is expected because the lack of knowledge in operating conditions when diagnosing such out domain scenarios, would mainly contribute to the epistemic uncertainty.

4.5. Results analysis for unseen noisy environments

The impact of noise through practical fault diagnosis is also important. The noise might be caused by external environment or instrumentation errors caused by sensor malfunctions. As listed in Table 1, the unseen noisy environments are simulated by adding additional

Gaussian noise to testing samples. Fig. 10 compares time waveforms of the testing samples in seen domain and 3 unseen noisy environments (25 dB, −5 dB and −35 dB). When the SNR falls below −5 dB, it is apparent that the fault signatures, such as periodic impact components, are totally submersed in the strong noise. Intuitively, it is challenging to identify the different bearing health conditions in the noisy environments of −5 dB and −35 dB.

Fig. 11 shows the diagnostic results for all the 400 testing samples and a sample of outer race fault sample as well as its predictions. According to Fig. 11(a), satisfactory diagnosis accuracies can be attained in the seen domain and the unseen noisy environments with high SNRs (over 10 dB), whereas significant degradation on accuracies can be found with SNRs less than −5 dB. This is consistent with the intuitive observations in Fig. 10. Moreover, the results in Fig. 11(b)–(d) are similar to the previous findings about unseen fault and unseen operating conditions, implying that probabilistic Bayesian deep learning has a superior capacity to identify out domain and provides trustworthy decision-making support.

Fig. 12 displays the uncertainty estimates and decomposition. It is observed that the total uncertainty is increasing given a higher noise level. When SNRs are less than −5 dB, a high noise is imposed and hence a significant rise in uncertainty. When SNRs are higher than 10 dB, the noise is relatively weak, and the uncertainty remains at a low level. From Fig. 12(i), there are no noticeable differences between the distributions of seen domain and the unseen noisy environments (40 dB and 25 dB). These findings, like those in the study of unseen fault and unseen operating conditions, demonstrate that uncertainty is effective in reflecting model trustworthiness of fault diagnosis in unseen noisy environments.

The results of uncertainty decomposition are presented in Fig. 12(a)–(h). Both the average of aleatoric and epistemic uncertainties follow an increasing trend given an higher noise level or a lower SNR as illustrated in Fig. 12(j)–(k). Most notably, aleatoric uncertainty becomes more prominent than epistemic uncertainty as the SNR decreases. This may contradict to the findings in Sections 4.3 and 4.4, and can be explained by the different nature of uncertainty sources. Particularly, aleatoric uncertainty, is referred to as the uncertainty inherent in the observed data. Once the input data is corrupted by too much noise, the fault diagnostic model becomes less confident with a large uncertainty, in which aleatoric uncertainty should take precedence over epistemic uncertainty.

4.6. Discussion

A comprehensive study of uncertainty-aware probabilistic Bayesian deep learning framework is investigated for machine fault diagnosis. The uncertainty estimation and decomposition offer various insights

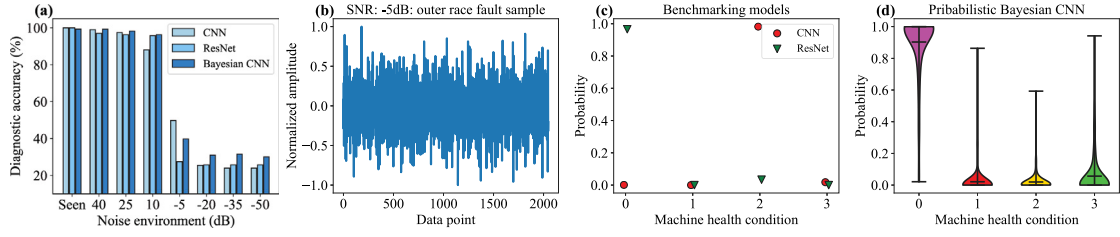


Fig. 11. The diagnosis of the testing samples of unseen noisy environments with benchmarking models and probabilistic Bayesian CNN. (a) Overall diagnostic accuracies under different noisy environments. (b) The original time waveform of a outer race fault sample under -5 dB noisy environments. (c)–(d) Diagnostic results of benchmarking models and probabilistic Bayesian CNN for the noisy sample.

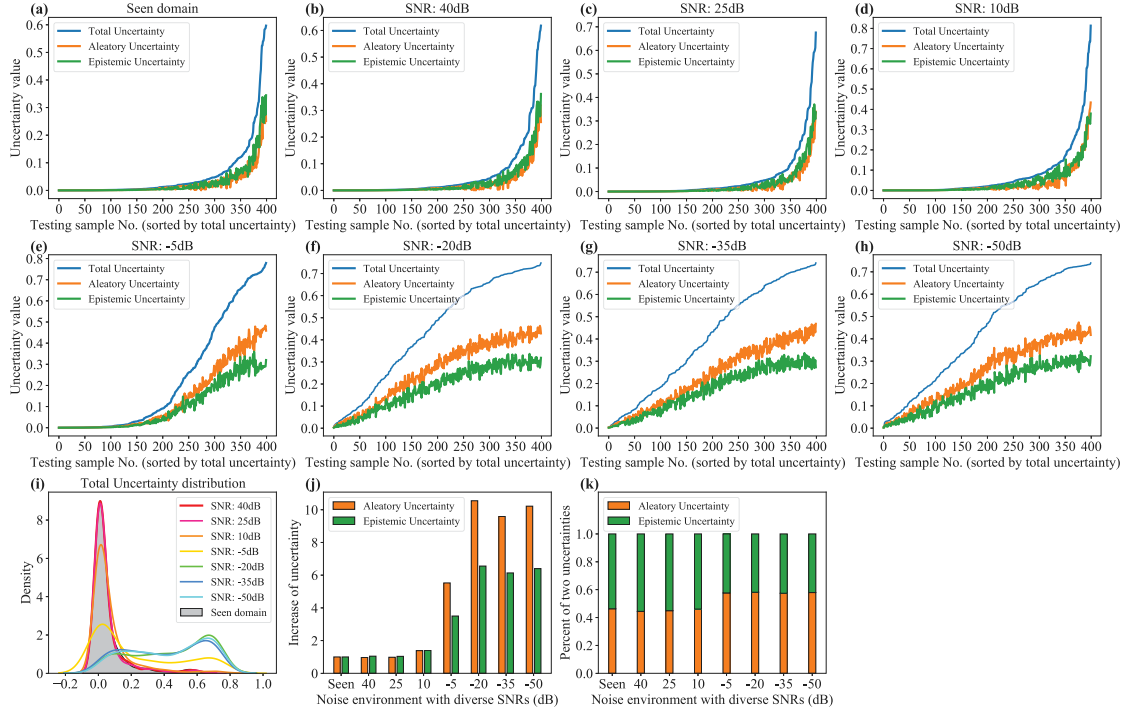


Fig. 12. Uncertainty estimation and decomposition for the testing samples of seen domain and different noisy environments. (a)–(h) The total uncertainty and decomposed uncertainty of testing samples in different domains. (i) The distribution of total uncertainty in different domains. (j)–(k) Analysis of uncertainty composition changes with respect to different domains.

in interpreting the diagnostic results, and issuing an alert for possible wrong diagnosis.

(1) The uncertainty estimation in probabilistic Bayesian deep learning framework is effective in forming the paradigm of trustworthy fault diagnosis. A high level of uncertainty indicates that the model is prone to making incorrect diagnosis and that the results are untrustworthy. The low uncertainty indicates that the input data are the same as or similar to the training data for developing the trained model, the model can be confident in its fault diagnosis. By delivering the uncertainty estimates to human experts, the transparency of these black-box diagnosis models can be improved, and the degree of trustworthiness of model's diagnosis can be quantified to a greater extent.

(2) Theoretically, epistemic uncertainty comes from the parameters of deep learning models, and represents the deviation of diagnostic results using different weights. Aleatoric uncertainty arises due to the noise in data measurement or unobserved factors, and reflects the inherent stochasticity in the diagnostic results with the fixed model weights. From the perspective of uncertainty decomposition, the important insights are listed below:

- In the diagnosis scenarios of unseen faults and unseen operating conditions, the data distribution of the target domain is generally different from the seen distribution during training. The domain

discrepancy exists in the label space of health conditions or the marginal distribution. Given a trained Bayesian deep learning model in advance, there is no adequate knowledge to diagnose such target data. As a result, the diagnostic outputs are uncertain with different weight samples drawn from the variational distribution, resulting in high epistemic uncertainty.

- In the diagnosis scenario of unseen noisy environments, additional noise will undoubtedly raise the randomness of raw data, and further disturb the results of fault diagnosis using our proposed method. Intuitively, as shown in Fig. 10, all the fault signature and diagnostic information are buried in the strong noise (-5 dB and -35 dB), suggesting the health conditions are not distinguishable in such noisy environments. The aleatoric uncertainty is simultaneously high in this scenario.
- Empirically, the testing samples far from the known distribution will induce the high epistemic uncertainty, while the inherent randomness of raw data hinders the confident diagnosis will result in high aleatoric uncertainty. The two sources of uncertainty illustrate different facets of the input data, and help us to understand the sources of uncertainty as well as the unknowns for a diagnostic model.

- The encouraging uncertainty decomposition informs the potential solutions for improving the diagnosis performance in unseen domains. To reduce the epistemic uncertainty, more data from the unseen domains (e.g. unseen faults or unseen operating conditions), are necessary, and can be obtained by additional experiment and simulation. The epistemic uncertainty vanishes in the limit of infinite data. The aleatoric uncertainty cannot be solved with more training data. To reduce it, we need to recur to more informative measurements about the noise or unobserved variables. Note that in practice, both of the additional fault data and measurements are often not available. In this case, we can only use the uncertainty decomposition to interpret the diagnostic decisions instead of improving the diagnosis model.

(3) In Section 4.4, the unseen rotating speeds are discussed to demonstrate the effectiveness of proposed method under unseen operating conditions. Note that the unseen operating conditions may include not only the unseen speeds, but also the unseen loads, or any other factors that may cause the domain shift. For the particular domain shift caused by speed changes, the order tracking is also a promising tool to deal with this challenge. By converting the non-stationary signals to angular domain, order tracking can collect the signal at multiple sampling frequencies depending on the machine's rotation frequency. It can thereby address the challenge of distribution shift induced by the change of speed conditions, and thus accurate diagnosis can be expected by using any deep learning architectures. The combination of order tracking and deep learning is worth investigating.

(4) In addition to the three aforementioned unseen scenarios, different fault degrees will lead to different dynamic responses in the mechanical system, and thus present different fault characteristics. In this context, the domain shift may also arise between seen fault degrees and unseen fault degrees, even for the same fault type. This factor has been discussed in several existing works [63–65]. Consequently, the changes of fault degrees may further influence the model performance for most of the data-driven methods. The impulsiveness of the time domain signal tends to be small with more severe fault levels. And the model trained on early fault cases may misdiagnose the progressed fault. Currently, intensive research about transfer learning, domain adaptation as well as generalization are conducted to solve it. This study provides a novel view to detect the unknown inputs and issue the untrustworthy diagnosis, which is a valuable supplement to existing data-driven fault diagnosis.

5. Conclusions

In this paper, we proposed a trustworthy machine fault diagnosis framework that improves the confidence in real-world applications. A probabilistic Bayesian CNN was developed to quantify predictive uncertainty considering both epistemic and aleatoric uncertainties. The uncertainty information can then be used to differentiate unseen domain. Our proposed framework can confidently diagnose faults in the seen domain with comparable performance to two benchmarking deep models, and it has a distinct capability in identifying unseen domain in practical fault diagnosis. Uncertainty decomposition is useful in distinguishing possible unseen domain scenarios caused by changes in fault type, rotating speed, and noise level. In general, this work can contribute to the field of machine fault diagnosis in two aspects: (1) facilitate the understanding of the trustworthiness of fault diagnostic results and inspire further research; (2) introduce a uncertainty-aware strategy for achieving trustworthy machine fault diagnosis in both seen and unseen domains.

Despite its strong and promising performance, trustworthy machine fault diagnosis is still in its infancy. Further research could include: assessing the effectiveness of the uncertainty-aware strategy in dealing with other unseen domain scenarios; developing more computationally efficient approaches for predictive uncertainty quantification; improving the trustworthiness of machine fault diagnosis by incorporating

physical knowledge of machine faults. Moreover, since the trustworthy machine fault diagnosis is a rarely explored topic, we need to firstly provide a unified and complete framework to investigate the uncertainty-aware trustworthy analysis and the uncertainty sources. All the analyses and comparisons are qualitative in this paper. To fulfill the automatic identification of untrustworthy diagnostic decisions and intelligently issue the warnings for human intervention, the next step should focus on the quantitative analysis of the trustworthiness of each testing sample. The proper determination strategies of uncertainty threshold need to be proposed so as to automatically define the untrustworthy diagnosis.

CRedit authorship contribution statement

Taotao Zhou: Writing – original draft, Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Te Han:** Writing – original draft, Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Enrique Lopez Droguett:** Validation, Investigation, Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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